

FT HOOD AAF, TX, USA

WMO: 722570

Lat: 31.133N

Lon: 97.717W

Elev: 282

StdP: 97.99

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.1	-17.4	0.8	4.6	-11.5	1.5	5.3	12.3	9.9	11.1	9.9	3.9	340	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	38.9	23.9	37.6	23.7	36.3	23.6	27.1	30.9	26.3	30.9	25.5	31.9	3.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.5	28.6	25.1	20.9	28.4	24.0	19.5	28.0	87.2	30.8	83.3	30.2	79.5	32.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.9	8.1	-6.1	40.1	1.8	1.9	-7.5	41.4	-8.5	42.5	-9.5	43.6	-10.9	45.0		
			-8.3	26.8	2.4	1.1	-10.0	27.5	-11.4	28.1	-12.8	28.7	-14.5	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.5	9.4	12.3	16.2	20.1	24.0	28.6	29.7	30.4	27.0	21.2	15.7	10.8
(7)		DBStd	8.43	5.48	6.27	5.30	4.16	3.42	2.28	1.94	2.26	2.98	4.65	5.13	5.35
(8)		HDD10.0	205	77	44	14	1	0	0	0	0	1	12	55	
(9)		HDD18.3	972	279	182	102	29	4	0	0	1	27	108	239	
(10)		CDD10.0	4037	60	109	207	303	434	557	611	632	511	348	183	81
(11)		CDD18.3	1762	4	14	36	82	181	307	353	373	261	116	29	7
(12)		CDH23.3	20397	31	109	251	678	1659	3776	4621	5096	2902	1048	186	40
(13)		CDH26.7	10355	3	26	52	196	651	1984	2586	2981	1446	393	35	4
(14)	Wind	WSAvg	3.9	3.9	4.3	4.6	4.7	4.2	4.0	3.7	3.3	3.1	3.4	3.6	3.6
(15)	Precipitation	PrecAvg	891	60	63	78	71	112	92	54	58	81	94	73	65
(16)		PrecMax	1211	143	189	232	150	273	246	156	204	214	280	256	225
(17)		PrecMin	550	7	2	0	12	27	13	0	2	9	11	7	5
(18)		PrecStd	201	39	46	47	40	58	57	41	59	54	69	63	55
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	30.3	30.6	33.4	35.4	39.3	39.9	41.3	38.9	34.1	29.6	26.9
(20)		0.4%	MCWB	16.1	15.7	18.2	21.2	23.8	23.8	23.9	24.5	22.9	21.5	20.4	17.9
(21)		2%	DB	23.4	26.8	28.0	30.8	33.6	37.2	38.4	39.4	36.5	32.0	27.6	24.2
(22)		2%	MCWB	14.7	14.9	18.5	20.5	23.1	24.0	23.7	24.3	22.7	21.1	19.4	17.2
(23)		5%	DB	20.9	24.1	26.1	28.9	31.9	35.7	37.0	37.8	34.8	30.7	25.5	22.0
(24)		5%	MCWB	13.7	16.2	18.1	20.2	22.6	24.0	23.6	23.8	22.7	20.7	18.5	16.0
(25)		10%	DB	18.7	21.8	24.3	27.3	30.4	34.4	35.7	36.5	33.4	29.0	23.5	19.9
(26)		10%	MCWB	12.5	15.3	17.2	19.4	22.2	24.0	23.6	23.7	22.5	20.1	17.5	14.9
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	20.3	21.6	24.1	26.4	28.0	28.2	26.7	25.8	26.5	22.5	20.6
(28)		0.4%	MCDWB	22.7	24.0	26.5	29.1	30.1	31.5	31.7	32.4	32.1	28.4	25.9	22.8
(29)		2%	WB	18.0	19.1	20.6	23.0	25.2	27.1	27.3	25.7	25.1	23.8	21.5	19.5
(30)		2%	MCDWB	21.4	22.3	24.4	27.8	29.5	31.0	30.7	32.9	31.2	28.2	24.9	22.6
(31)		5%	WB	16.1	18.1	19.9	22.1	24.5	26.2	26.3	25.2	24.5	23.0	20.5	18.0
(32)		5%	MCDWB	19.4	21.4	23.6	26.4	29.1	30.9	30.5	33.3	30.5	27.7	23.7	20.6
(33)		10%	WB	13.0	16.7	19.1	21.3	23.8	25.2	25.3	24.8	24.0	22.3	19.3	16.2
(34)		10%	MCDWB	17.1	20.8	22.7	25.5	28.4	31.4	32.5	32.7	29.8	26.6	22.7	18.9
(35)	Mean Daily Temperature Range	MDBR	11.5	11.7	11.6	11.5	10.3	10.9	11.3	11.7	11.3	11.8	11.3	11.1	
(36)		5% DB	MCDBR	15.2	13.9	13.0	12.6	11.9	12.4	12.7	13.0	12.8	12.8	12.3	13.7
(37)		5% DB	MCWBR	8.8	7.7	6.1	6.0	5.1	3.4	2.9	3.1	3.7	5.3	6.5	8.3
(38)		5% WB	MCDBR	12.2	11.2	10.5	10.6	10.1	10.9	10.8	11.7	10.5	10.2	10.0	11.3
(39)		5% WB	MCWBR	9.0	8.0	6.3	6.2	4.7	3.3	3.6	3.2	3.7	5.2	6.7	9.0
(40)	Clear-Sky Solar Irradiance	taub	0.313	0.327	0.353	0.393	0.427	0.423	0.435	0.426	0.396	0.367	0.342	0.317	
(41)		taud	2.519	2.482	2.423	2.333	2.294	2.345	2.346	2.356	2.388	2.453	2.468	2.511	
(42)		Ebn at Noon	919	935	930	899	866	865	852	858	874	878	875	894	
(43)		Edh at Noon	86	98	112	127	133	126	126	123	115	100	91	83	
(44)	All-Sky Solar Radiation	RadAvg	3.02	3.52	4.56	5.48	5.81	6.64	6.74	6.27	5.22	4.20	3.17	2.66	
(45)		RadStd	0.32	0.59	0.40	0.50	0.53	0.51	0.51	0.36	0.45	0.62	0.38	0.36	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71

Nomenclature: See separate page